

Graph-Based Retrieval Enhancement for Semi-Structured Data



Authors:

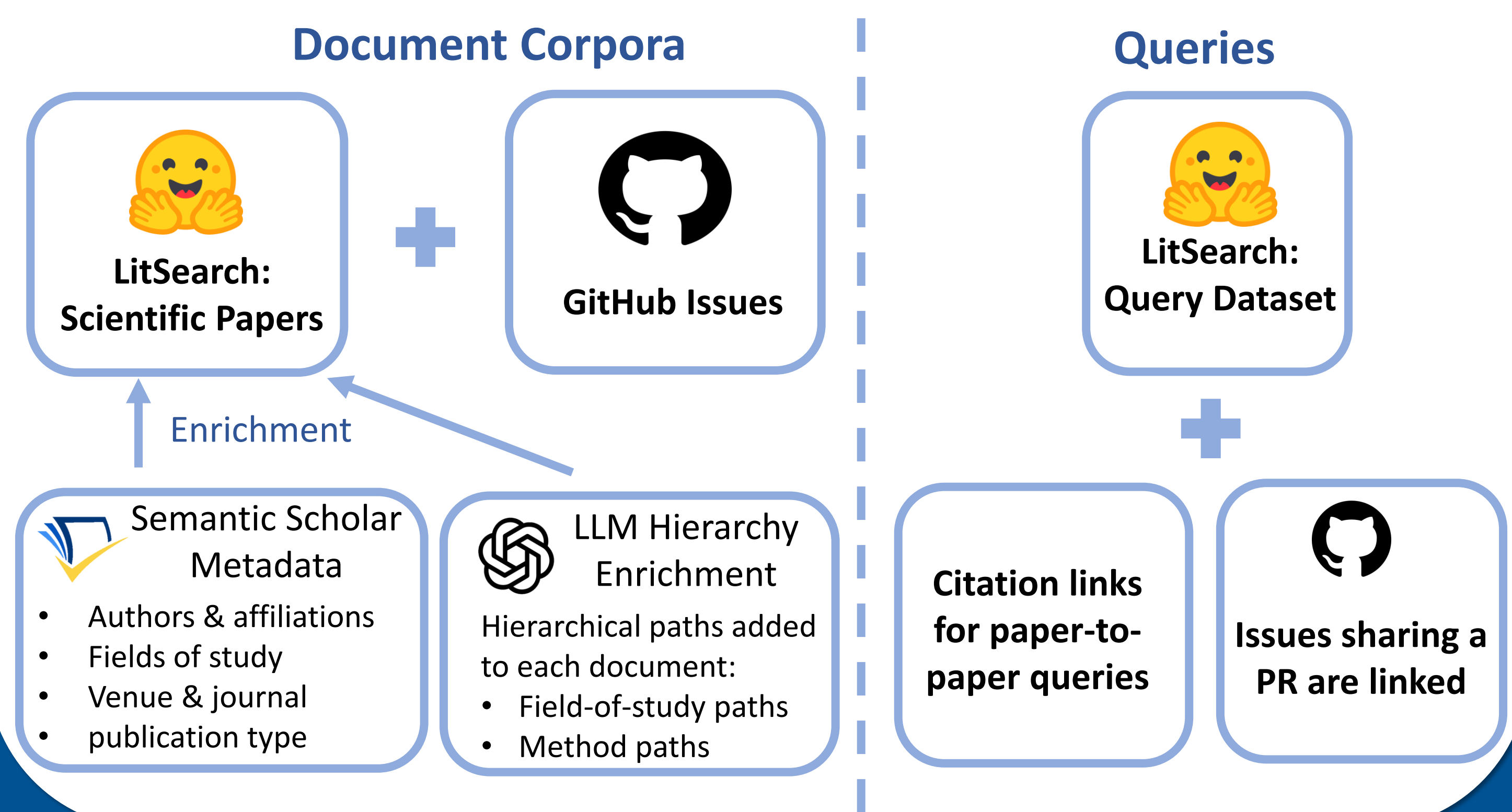
Olga Czajkowska, Michaela Blomeier, Dennis Halilaj

Project Supervisor:

Jakob Sturm

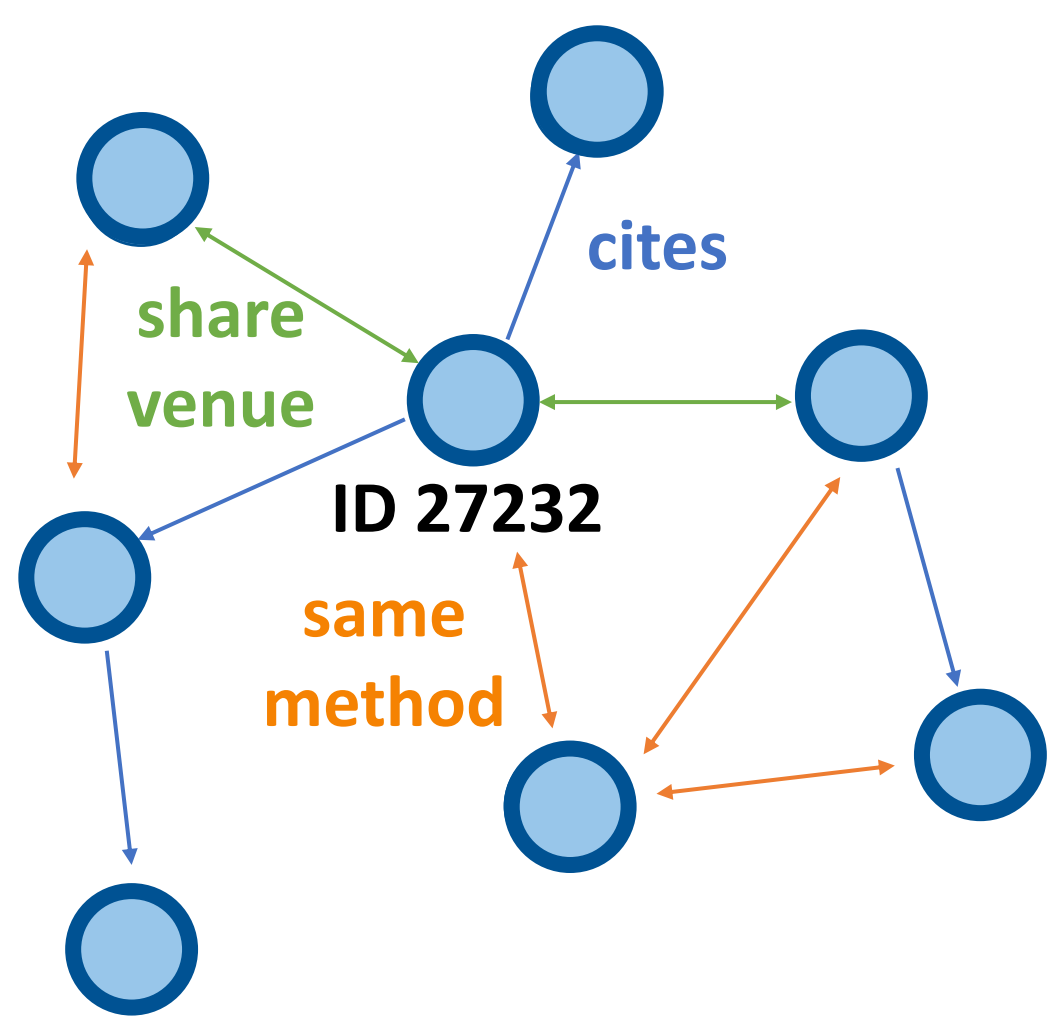
Can retrieval be improved by exploiting structural relationships between documents rather than treating them as isolated text?

Datasets and Enrichment



Data Representation & Graph Structure

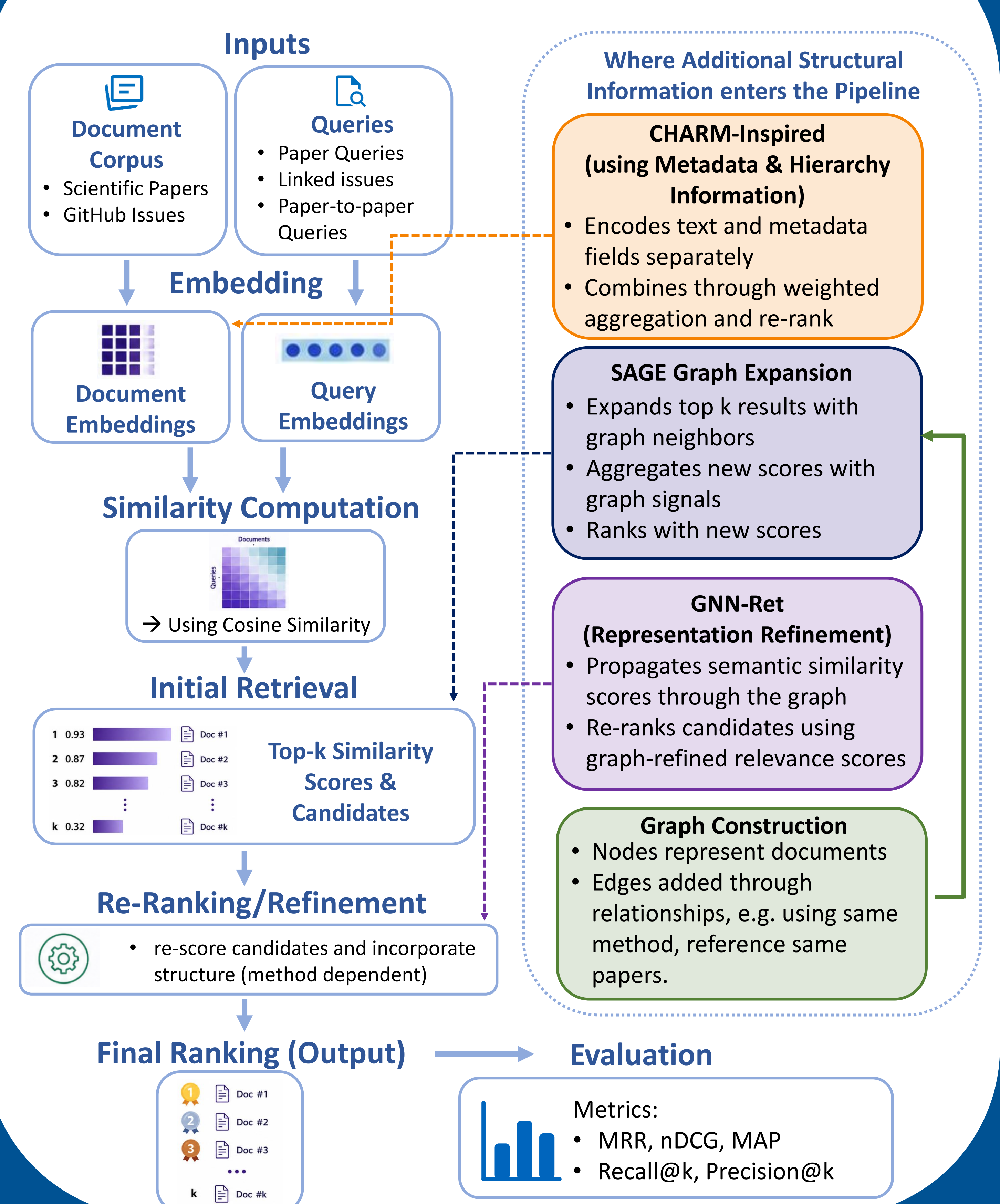
Example of Graph Structure



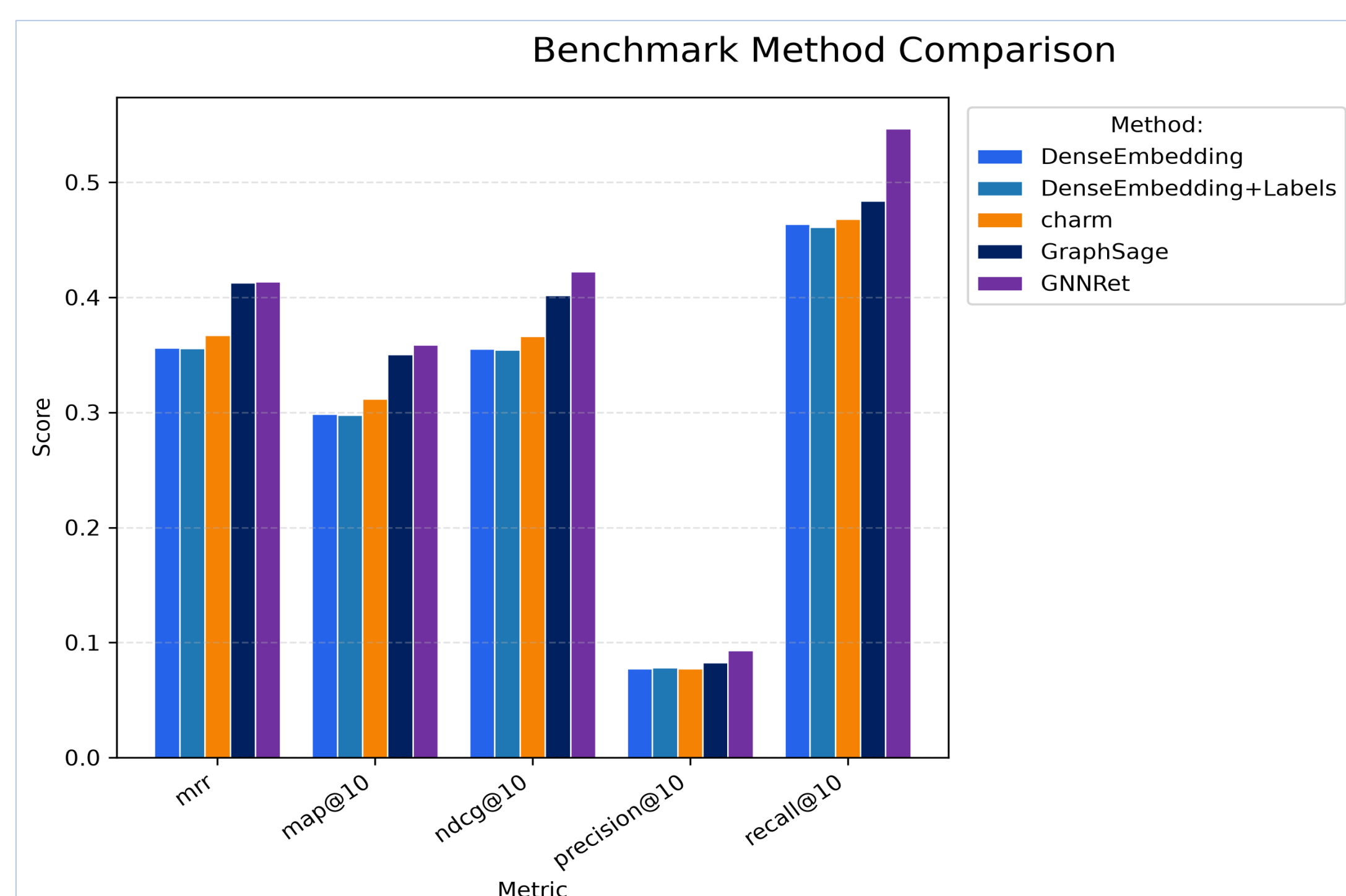
Example Document

ID 27232
Source Semantic Scholar
Title
PHENAKI: Variable Length Video Generation from Open Domain Textual Descriptions
Metadata
Venue: ICLR
Organizations: Universidad de la República, University of Michigan
Hierarchy
Deep Learning → Transformer based generation → ...
Relations
3 connected documents

Pipeline Overview



Benchmark Results



BASELINES

BM25

Lexical term matching is strong, but wording sensitive baseline

Dense Retrieval

Semantic embeddings improves retrieval beyond term overlap

STRUCTURE-AWARE METHODS

CHARM-Inspired

Weighted metadata & hierarchy embedding fields provide limited gains over dense retrieval

SAGE Graph Expansion

Graph-neighbor expansion improves retrieval when structural links align with relevance

GNN-Ret

Adding a GNN-based refinement layer improves the retrieval

Conclusions

Graph Structure provides the strongest retrieval gains

Including document relations can recover relevant candidates missed by text-only similarity

The value of structure depends on the retrieval task

Gains are strongest when graph relations, such as citations or linked issues align with relevance

Metadata alone is not sufficient

Encoding categorical and hierarchical metadata as weighted text fields yields limited improvement